

Reflection on Software Quality: Comparing Verdugo et al. (2024) and Hanief et al. (2025)

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Verdugo et al. (2024) trace a *25-year research–practice journey* focused on *software product quality evaluation and certification*. The article emphasises the transfer of **software quality metrics research into industrial use**, culminating in the creation of an **accredited laboratory** that evaluates products against international standards (ISO/IEC 25000 family) with particular focus on **functional suitability** and **maintainability**. Its central contribution lies in *bridging academic definitions and industrial evaluation practices*, making quality evaluation rigorous, repeatable, and *certifiable* within a standards-based framework, thus enabling organisations to *measure and certify* their products against globally recognised quality criteria. The article further highlights the historical evolution of metrics, frameworks and empirical validations used to support quality evaluation, indicating the *maturity and methodological depth* in how quality attributes are assessed in practice (Verdugo et al., 2024).

In contrast, the 2025 *Systematic Literature Review on Software Quality Using Agile Approach* by Hanief et al. adopts a *literature review methodology* to understand how **software quality is measured and managed in Agile environments**. Rather than focusing on certification or standards application, this study examines *empirical evidence* about metrics and practices specific to Agile’s iterative model, including **sprint velocity, defect density, and test coverage**. It identifies the strengths and risks associated with Agile quality metrics, such as the potential for *technical debt when velocity is used in isolation*, emphasising the need for *internal code quality measures* like code churn and cyclomatic complexity alongside automated testing and **shift-left strategies**. Furthermore, the Agile SLR points to *emerging holistic frameworks* integrating quantitative and qualitative assessment, supported by AI-assisted testing and predictive analytics in CI/CD pipelines. This sets its focus on *contextual, evolutionary quality practices* rather than formal product certification.

The key difference between the two articles lies in their **scope and purpose**. Verdugo et al.'s work is centred on **formal quality evaluation and certification**, grounded in international standards and the translation of research into *industrial evaluation artefacts*. In contrast, Hanief et al. (2025) emphasise **Agile-specific quality measurement and management practices**, shaped by iterative delivery, team practices, and tooling. Where Verdugo et al. treat quality as a *certifiable property of a product*, the Agile SLR treats quality as a *process and metric-driven outcome* of iterative software development. Together, these perspectives highlight that software quality can be understood both through *standards-aligned evaluation frameworks* and *context-sensitive Agile quality practices*, depending on project goals and organisational context.

References

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